

5.1 Working with Coordinates

- Midpoint $= \left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$
- Magnitude between 2 points $\Rightarrow |AI| = \sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$
- Gradient: $m = \frac{\Delta y}{\Delta x} = \frac{y_2-y_1}{x_2-x_1}$

Parallel: $m_1 = m_2$ Perpendicular: $m_1 \cdot m_2 = -1$

Equation of a line: $y - y_1 = m(x - x_1)$

If given a gradient & a point, directly substitute

y-intercept, c, $(x_1, y_1) = (0, c)$

If you are given 2 point, find m, then substitute in one point

$(x_1, y_1) = (4, 3)$ $y - 3 = 2(x - 4)$

$m = 2$

Equations are generally left in $y = mx + c$ / $ax + by + c = 0$

5.2: Intersection of lines:

In 2D space, lines are either parallel or they intersect.

To find an intersection, both graph conditions must be kept simultaneously, point must obey both equations. Therefore use sim. equations.

5.4: The Circle:

Equation of circle with centre (a,b) is $(x-a)^2+(y-b)^2=r^2$

(Think of it as subtracting the center to go back to the origin, from there you get the exact radius, without it would add center, and still go from 0,0)

If $C=(0,0) \Rightarrow x^2+y^2=r^2$

By completing the square for x and y components of the expanded form of the circle $x^2+y^2+ax+by+c=0$ it can be converted into the original form

Theorems:
Angle in semi-circle is a right angle.
The perpendicular from the center to a chord bisects a chord
Tangent is perpendicular to 'r' at that point

5.5: Intersection of line and curve.

Must also solve simultaneously.

1) all intersections are distinct.

2) The line is a tangent to the curve. The point of contact corresponds to 2 intersections. It may also intersect elsewhere. FOR A QUADRATIC.

3) There is no intersection